Class 10 Math - Chapter 9: Some Applications of Trigonometry

🌟 Heights and Distances

This chapter demonstrates how the theoretical ratios of trigonometry (learned in Chapter 8) are applied strictly in the real world to measure towering structures, vast rivers, and flying objects without ever using a physical measuring tape.

In every problem, you will mentally translate a real-world scenario directly into a **Right-Angled Triangle**.

📌 Angles of Elevation and Depression

1. Angle of Elevation

When an observer looks upwards at an object above their horizontal eye line.

- **Line of Sight:** The imaginary line drawn from the eye of the observer directly to the object.
- **Formation:** The angle formed *above* the horizontal line.

2. Angle of Depression

When an observer looks downwards at an object below their horizontal eye line.

- **Formation:** The angle formed *below* the horizontal line.

> [!NOTE] Because the horizontal eye line is parallel to the ground, the **Angle of Depression** from the top is structurally precisely equal to the **Angle of Elevation** from the ground due to the geometric property of *Alternate Interior Angles*!

Mind Map of Sight Lines

```
graph TD;
    Observer_Eye-->Horizontal_Line;
    Horizontal_Line-->Looks_Up["Looks Up\n(Angle of Elevation)"];
    Horizontal_Line-->Looks_Down["Looks Down\n(Angle of Depression)"];
    Looks_Up-->Target_Above;
    Looks_Down-->Target_Below;
```

⚡ Solving Strategy

To conquer any height and distance problem:

1. **Draw a Figure:** Accurately sketch the scenario using a simple right-angled triangle. Label the right angle, the height, the base, and the given angle.
2. **Identify Knowns and Unknowns:** What side do you have? What side do you want?
3. **Choose the Ratio:** Which ratio connects the known and the unknown?
   - Need P but have H? Use $\sin \theta$.
   - Need B but have H? Use $\cos \theta$.
   - Need H but have P? Use $\tan \theta$ (actually most problems use $\tan$ since structures are strictly perpendicular to the flat ground base).

> [!IMPORTANT] Because buildings, trees, and towers stand vertically directly on the ground, always assume the angle between the structure and the ground is exactly 90°.

❌ Common Mistakes

> [!WARNING]

> 1. **Ignoring Observer Height:** If the problem explicitly states "a boy 1.5m tall is looking at a building...", you MUST draw the horizontal line of sight 1.5m above the ground. You calculate the top part of the building using trigonometry, then mathematically add 1.5m to get the building's total height!
> 2. **Placing Depression Angle inside:** Students often place the angle of depression dynamically inside the triangle next to the vertical wall. The angle of depression is formed with the HORIZONTAL air line extending outwards, not the vertical building wall.
> 3. **Using wrong standard values:** Swapping $\tan 30^\circ$ ($1/\sqrt{3}$) and $\tan 60^\circ$ ($\sqrt{3}$) will instantly ruin your calculation.

📝 Quick Revision

The Golden Rule of $\tan$

In $90%$ of height and distance problems, you are given the distance from the base (Base) and asked for the height of the tower (Perpendicular). Thus: $$ \tan \theta = \frac{\text{Height}}{\text{Distance}} \Rightarrow \text{Height} = \text{Distance} \times \tan \theta $$

Summary

Applications of trigonometry transform complex visual word problems into simple geometric sketches. Draw an accurate diagram, properly assign your elevation or depression angles, select the correct SOH-CAH-TOA relation, and plug in your standard angle exact values to find the requested line distance!